

EFFINGHAM, IL, USA

WMO: 744661

Lat: 39.070N

Lon: 88.534W

Elev: 179

StdP: 99.19

Time Zone: -6.00 (NAC)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.1	-12.3	-20.2	0.6	-13.8	-17.5	0.8	-11.4	11.4	1.8	10.4	-1.4	2.8	290	0.447

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	34.0	23.9	32.7	23.7	31.5	23.4	26.1	31.1	25.3	30.3	24.6	29.3	3.3	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.8	20.3	29.0	23.9	19.2	27.9	23.0	18.1	27.1	82.2	31.4	78.5	30.2	75.2	29.3	36.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.3	8.2	7.3	DB	-18.9	36.6	3.4	2.8	-21.3	38.6	-23.3	40.2	-25.2	41.7	-27.6	43.7	(4)
(5)			WB	-19.5	27.7	3.3	2.5	-21.8	29.5	-23.7	31.0	-25.6	32.4	-27.9	34.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.8	-1.3	0.5	7.1	13.0	18.5	23.3	24.8	23.9	20.4	13.7	6.8	1.7
	DBStd	10.45	6.54	6.34	6.24	5.17	4.69	3.16	2.97	2.90	3.94	5.37	5.45	5.50
	HDD10.0	1200	354	272	134	28	2	0	0	0	0	22	126	263
	HDD18.3	2745	609	499	352	169	59	3	1	2	23	165	347	516
	CDD10.0	2207	3	6	43	119	264	398	457	431	313	136	31	5
	CDD18.3	709	0	0	3	10	63	152	199	174	86	21	1	0
	CDH23.3	6778	0	0	17	104	544	1447	2002	1608	848	204	6	0
	CDH26.7	2576	0	0	1	13	153	555	846	627	322	61	0	0
Wind	WSAvg	2.9	3.8	3.6	3.6	3.7	2.8	2.3	1.9	1.7	1.9	2.7	3.2	3.4
Precipitation	PrecAvg	995	53	57	95	90	100	105	104	68	79	67	89	85
	PrecMax	1340	140	130	195	196	205	310	244	169	219	197	263	251
	PrecMin	697	1	11	23	23	24	15	18	11	5	11	10	8
	PrecStd	169	38	31	48	46	47	64	58	41	58	39	55	57
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.7	19.7	26.0	27.9	32.1	34.5	37.3	36.0	34.1	31.4	23.7	17.9
		MCWB	15.1	15.1	17.3	18.0	22.3	23.8	23.8	24.8	22.0	22.3	16.6	14.6
	2%	DB	14.1	16.4	22.5	26.3	30.0	32.8	34.1	33.3	32.4	27.9	20.9	15.1
		MCWB	11.9	13.6	16.0	17.3	21.3	23.3	24.6	24.2	22.1	20.4	15.8	13.2
	5%	DB	12.0	13.1	19.8	24.0	28.4	31.7	32.6	32.0	30.6	25.5	17.9	12.6
		MCWB	9.8	10.2	14.4	16.4	20.6	23.1	24.6	24.2	21.7	18.2	14.1	10.3
	10%	DB	8.0	10.1	17.4	22.3	27.0	30.0	31.2	30.1	28.0	22.8	16.0	10.1
		MCWB	6.1	7.6	13.5	15.6	19.6	22.1	23.9	23.2	20.2	17.0	12.0	7.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.6	16.5	18.4	20.4	24.1	26.6	27.4	27.3	25.0	23.3	19.1	15.7
		MCDB	17.3	18.4	23.5	25.6	28.8	31.7	32.3	32.6	29.4	28.0	22.7	17.2
	2%	WB	12.9	13.9	16.9	18.9	22.7	25.2	26.2	25.8	23.8	21.5	16.9	13.4
		MCDB	13.9	16.0	21.1	23.6	27.6	30.4	31.4	30.6	28.7	26.2	19.1	14.9
	5%	WB	9.9	10.7	15.6	17.7	21.7	24.1	25.5	25.0	22.8	20.1	15.0	10.6
		MCDB	11.3	12.8	18.9	22.2	26.5	29.0	30.5	29.8	27.7	23.9	17.9	12.1
	10%	WB	6.0	7.5	13.9	16.6	20.7	23.3	24.8	24.2	21.8	18.2	12.1	7.9
		MCDB	8.3	10.1	17.0	21.1	25.2	28.2	29.6	28.5	26.6	21.6	15.0	9.3
Mean Daily Temperature Range	5% DB	MDBR	8.5	8.9	10.1	11.3	10.8	11.0	10.7	11.0	12.7	11.9	9.8	7.8
		MCDBR	12.2	13.1	13.1	13.8	12.5	12.5	12.0	12.5	14.6	14.3	12.9	10.6
		MCWBR	10.2	10.3	8.0	7.4	5.6	5.2	4.6	5.2	5.6	6.8	8.5	8.7
	5% WB	MCDBR	11.3	12.7	11.1	11.9	10.6	10.7	10.6	10.8	12.1	11.6	11.3	9.9
		MCWBR	10.5	10.8	7.6	7.5	5.4	5.2	4.8	5.3	5.5	6.3	8.7	9.1
Clear-Sky Solar Irradiance	taub	0.305	0.321	0.356	0.399	0.436	0.436	0.462	0.462	0.415	0.360	0.317	0.304	
	taud	2.456	2.396	2.370	2.248	2.200	2.233	2.155	2.144	2.265	2.417	2.488	2.474	
	Ebn at Noon	871	902	900	877	848	846	819	808	827	845	849	844	
	Edh at Noon	80	98	111	134	144	139	149	147	123	95	78	74	
All-Sky Solar Radiation	RadAvg	1.88	2.69	3.68	4.85	5.49	6.35	6.18	5.57	4.67	3.32	2.23	1.59	
	RadStd	0.20	0.28	0.33	0.29	0.48	0.41	0.39	0.34	0.33	0.35	0.24	0.18	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (38 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+78

Nomenclature: See separate page